

What does your pressure space miss?

Compare a prescribed log-volume field, its pressure-space projection, and the component that the space cannot represent.

A representation lesson, at a fixed state.

This is a finite weighted example. There is no body deformation, equilibrium solve, or incompressibility simulation. Refinement changes the space, while the prescribed field stays the same.

Full sample energy

8

$$E_{\text{full}} = K/2 \sum w_i g_i^2$$

Condensed energy

0.5

$$E_{\text{cond}} = K/2 \sum w_i (Pg)_i^2$$

Unresolved energy gap

7.5

$$E_{\text{full}} - E_{\text{cond}} = K/2 \sum w_i r_i^2$$

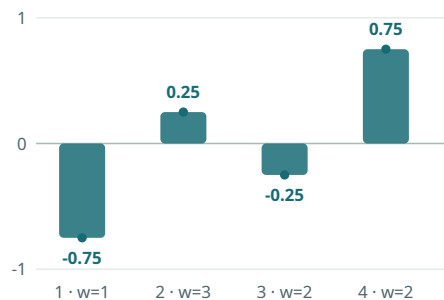
The field, its projection, and the remainder

Discrete samples · one shared vertical scale · no material geometry

Prescribed field

g

Authored log-volume samples

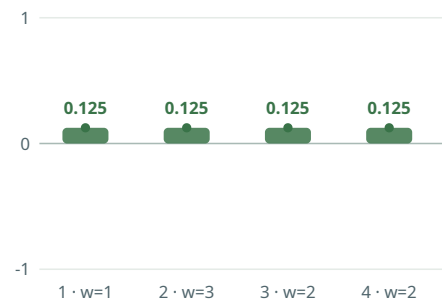


[-0.75, 0.25, -0.25, 0.75]

Represented part

Pg

One weighted mean, repeated

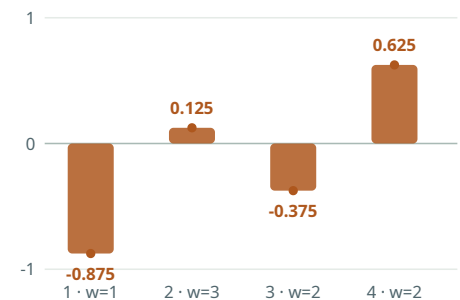


[0.125, 0.125, 0.125, 0.125]

Unresolved part

r

$r = g - Pg$



[-0.875, 0.125, -0.375, 0.625]

Actual values used in this calculation

Sample	w _i	g _i	(Pg) _i	r _i	p [*] _i = K(Pg) _i
1	1	-0.75	0.125	-0.875	1
2	3	0.25	0.125	0.125	1
3	2	-0.25	0.125	-0.375	1
4	2	0.75	0.125	0.625	1

p^{*} is the optimal mixed variable in the chosen sample space, not a measured anatomical pressure.

Full: 4 × 2 = 8

Condensed: 4 × 0.125 = 0.5

Residual: 4 × 1.875 = 7.5

Mixed objective at p^{*}: 0.5

Readouts round to five decimal places; the identity diagnostic uses unrounded values.

Weighted residual sums by group: [0]. Browser identity difference: 0.

What refinement recovers

Every row uses the same current g, weights, and K.

Pressure values	Econd	Gap
● 1 shared	0.5	7.5
● 2 groups	1	7
● 4 samples	8	0

Refining from one shared value to two groups recovers the pair means. Four sample values recover this entire prescribed field, with zero gap.

Energy ordering is at a fixed state. It does not establish stability or compare solutions re-equilibrated in different spaces.

THE PROJECTION IS EXPLICIT

$$(Pg)_i = \sum_{j \in G_i} w_j g_j / \sum_{j \in G_i} w_j$$

G_i is the selected group containing sample i. Q consists exactly of vectors constant on each group. Positive weights make each group denominator positive.

WHY THIS IS ORTHOGONAL

$$\sum_{j \in G} w_j (g_j - (Pg)_j) = 0$$

The unresolved part has zero weighted sum in each group. It is therefore weighted-orthogonal to every group-constant vector in Q. Ordinary unweighted averages would give a different projector.

Exact algebra and browser arithmetic are separate.

The 27 Lean statements prove weighted projection, unique maximizing pressure vectors, condensed energy, the nonnegative gap, and nested-space monotonicity under exact assumptions. This lab checks its numerical implementation separately.

[Read the standalone proof guide and exact reference](#) · [Lean source](#)

Interpretation and limits